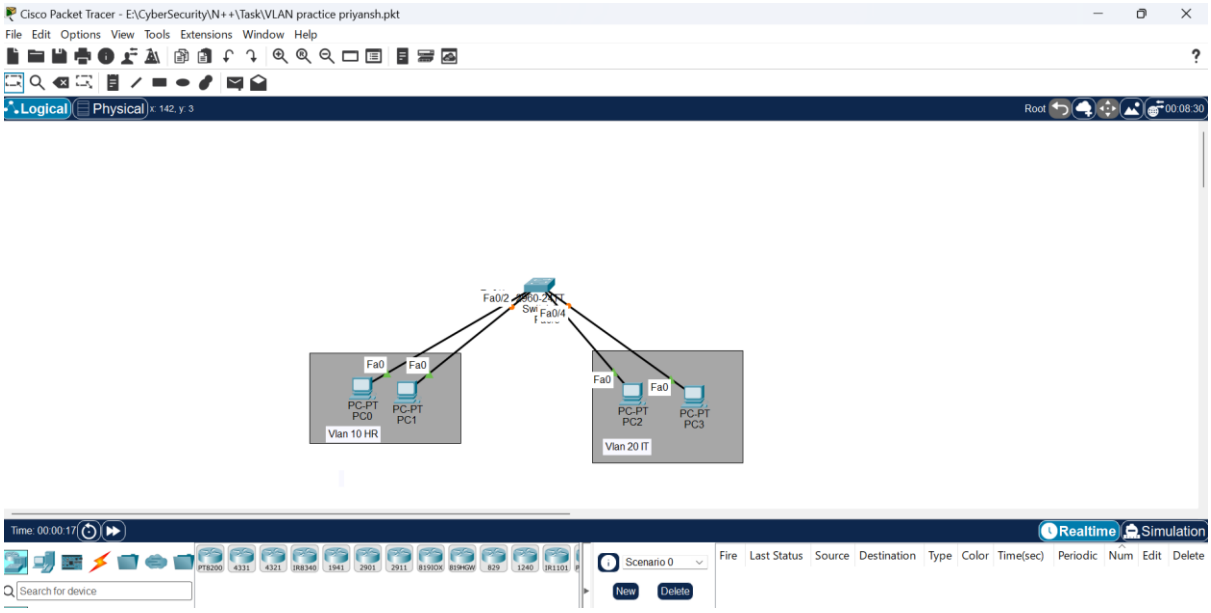
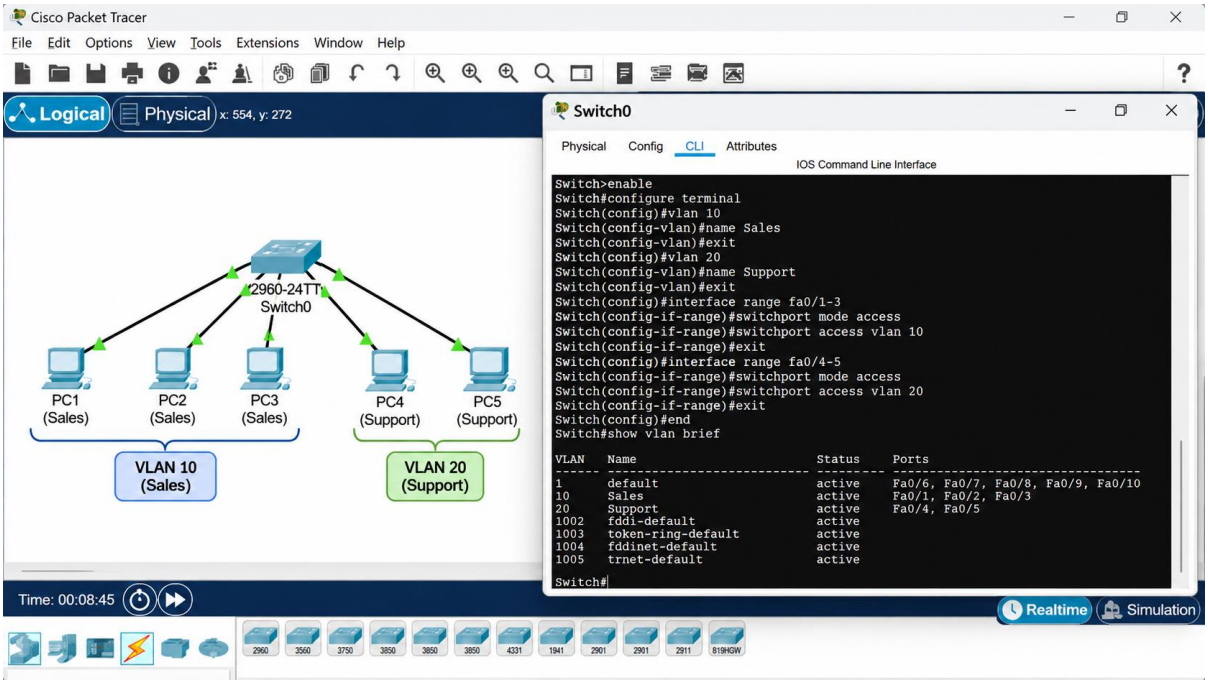
Day3 Switch and Router Config

Task 1:- Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.



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Task 2:- Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Use the interface range command to select multiple ports efficiently.



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Task 3:- Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

step 1: Create the Network Topology

Open Cisco Packet Tracer and add:

- 2 Routers (1941)
- 2 Switches (2960)
- 2 PCs (PC0 and PC1)

Connect them using Ethernet cables (Copper Straight-Through for PC-to-switch and switch-to-router; use Copper Cross-Over or automatic cable selection between routers).

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Step 2: Assign IP Addresses

Use the following IP addressing plan:

Device	Interface	IP Address	Subnet Mask
Router 1	G0/0	192.168.1.1	255.255.255.0
Router 1	G0/1	10.0.0.1	255.255.255.252
Router 2	G0/0	10.0.0.2	255.255.255.252
Router 2	G0/1	192.168.2.1	255.255.255.0
PC0	FastEthernet	192.168.1.2	255.255.255.0
PC1	FastEthernet	192.168.2.2	255.255.255.0

Default gateways:

- PC0: `192.168.1.1`
- PC1: `192.168.2.1`

Step 3: Configure Router 1

Click Router 1 → CLI and enter:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252
no shutdown
exit
```

Step 4: Configure Router 2

Click Router 2 → CLI and enter:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

Step 5: Configure Static Route on Router 1

Router 1 needs to know how to reach the LAN behind Router 2.

Enter these commands on Router 1:

```
enable
configure terminal

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end
```

This tells Router 1 to forward traffic for `192.168.2.0/24` through Router 2 at `10.0.0.2`.

For two-way communication, also configure a return route on Router 2:

```
enable
configure terminal

ip route 192.168.1.0 255.255.255.0 10.0.0.1

end
```

Step 6: Verify the Configuration

On Router 1, enter:

```
show ip route
```

You should see a static route marked with `S` for `192.168.2.0/24`.

Test connectivity from PC0:

```
ping 192.168.2.2
```

Expected result

Successful replies indicate connectivity between the two LANs.

Task 4:- Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Think about router-on-a-stick or Layer 3 switch concepts.

Inter-VLAN Communication: Sales VLAN and Support VLAN

1. Scenario:

- VLAN 10: Sales – 192.168.10.0/24
- VLAN 20: Support – 192.168.20.0/24
- PC1 (Sales): 192.168.10.2
- PC2 (Support): 192.168.20.2

2. Ping Test:

From PC1, run:

```
ping 192.168.20.2
```

Result: Ping fails and shows `Request timed out` because both PCs belong to different VLANs.

3. Reason for Failure:

- VLANs are separate logical networks.
- A Layer 2 switch cannot route traffic between different VLANs.
- Inter-VLAN routing is required for communication.

4. Solution: Router-on-a-Stick

Configure subinterfaces on the router:

```
enable
configure terminal

interface gigabitEthernet 0/0
no shutdown
exit

interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

Configure the switch trunk port:

```
interface fa0/24
switchport mode trunk
exit
```

5. Default Gateway Configuration:

- PC1 (Sales): 192.168.10.1
- PC2 (Support): 192.168.20.1

6. Verify Connectivity:

Run the ping command again:

```
ping 192.168.20.2
```